

HealthConnect Clinic

Project Summary

1. Project Overview

Week 6 extended the HealthConnect analytics work developed in the previous week by moving from descriptive exploration toward deeper relationship analysis, validation of key findings, and decision support insights. The focus was on understanding factors associated with appointment no-shows and presenting those relationships in a clearer analytical dashboard.

2. Objectives

- Built on the existing Week 5 dashboard and exploratory analysis.
- Examine relationships between booking behaviour, previous no-show history, and appointment outcomes.
- Compare no-show patterns across relevant patient and appointment segments.
- Validate and refine findings from the previous analysis.
- Identify high impact patterns that can support operational decision-making.
- Improve the Power BI dashboard to communicate deeper insights effectively.

3. Analytical Work Completed

The analysis was extended through additional grouping, segmentation, and relationship analysis. A Previous No-Show Group was created from the existing `previous_no_shows` field, grouping records into 0, 1, 2, and 3+ previous no-shows. This enabled clearer comparison of patients according to their previous appointment history.

- Analysed No-Show Rate by Booking Lead Group.
- Analysed No-Show Rate by Previous No-Show Group.
- Compared Booking Lead Group against Previous No-Show Group using a matrix.
- Compared Previous No-Show Group against Reminder Channel using a matrix.
- Used Total Appointments alongside rates where appropriate to provide context for segment size.
- Applied conditional formatting to analytical matrices to make higher-risk segments easier to identify.
- Reviewed KPI and dashboard findings against the earlier analysis to support validation and refinement.

4. Dashboard Development

The Week 6 Power BI dashboard was developed as an extension of the existing HealthConnect work rather than a replacement of the earlier dashboard. The updated dashboard was structured to separate the main decision-support view from more detailed analytical views.

- Page 1 was developed as the main decision-support view, incorporating KPI cards, line charts, a table, and a combined no-show risk analysis.
- Page 2 was developed as an analytical detail page containing clustered column charts, matrices, and tables.
- The dashboard incorporated the new Booking Lead Group and Previous No-Show Group analysis.
- Analytical matrices were formatted to highlight differences in no-show rates.
- Dashboard improvements were designed to make important risk patterns easier to interpret and communicate.

5. Key Findings

The Week 6 analysis strengthened the understanding that appointment no-show behaviour is associated with more than one factor. Booking lead time and previous no-show history emerged as important analytical dimensions for identifying higher-risk segments.

- No-show rates increased across the booking lead groups, with appointments booked further in advance showing higher observed no-show rates.
- Patients with a history of previous no-shows generally showed higher current no-show rates than patients with no previous no-shows.
- The combined Booking Lead Group $\times$ Previous No-Show Group analysis provided a more useful risk view than examining either variable independently.

6. Business Impact and Decision Support

The analysis provides a basis for prioritising appointment-management efforts. Rather than applying the same intervention to every appointment, the identified patterns can support a more targeted approach. Patients or appointments showing combinations of higher-risk characteristics could be considered for additional monitoring, reminder strategies, or other operational interventions.

- Use booking lead time as a risk-monitoring dimension when planning appointment follow-up.
- Pay particular attention to patients with repeated previous no-shows.
- Consider combining multiple risk indicators when prioritising follow-up rather than relying on one variable.
- Use reminder channel performance as an area for further testing and optimisation.
- Monitor segment volumes before making operational decisions based on very high or very low rates.

7. Cross-Track Collaboration and Data Science Integration

Cross-track collaboration with a Data Science intern was attempted during Week 6; however, I wasn't able to collaborate with a Data Science intern during the period. As a result, formal cross-track implementation or model development could not be completed within this week's work.

Although the collaboration was not completed, the analytical work was structured to provide a useful handoff for future Data Science integration. The findings identify potential variables and relationships that could be evaluated in a predictive no-show model.

- Booking lead days / Booking Lead Group as a potential predictive feature.
- Previous no-shows / Previous No-Show Group as a potential predictive feature.
- Reminder channel as a potential explanatory or predictive variable.
- The interaction between booking lead time and previous no-show history as a potential model feature or interaction term.
- Actual appointment outcomes as the basis for comparing predicted and observed no-show behaviour.

8. Limitations and Assumptions

- The analysis identifies observed relationships in the available data and does not establish causation.
- Differences in rates should be interpreted together with the number of appointments in each segment.
- Small groups may produce unstable or unusually high rates.
- Reminder-channel differences should not automatically be interpreted as evidence that one channel causes better attendance.
- Cross-track predictive modelling was not completed because a Data Science collaborator was unavailable during Week 6.
- Further validation is required before using the identified factors as a formal operational risk-scoring system.

9. Week 7 Readiness and Testing Focus

The Week 6 analysis provides a foundation for testing whether the identified patterns remain reliable when evaluated more systematically. Future testing should focus on the predictive usefulness of booking lead time, previous no-show history, reminder channel, and their combinations.

- Test whether booking lead time consistently improves no-show risk identification.
- Test whether previous no-show history adds meaningful predictive information.
- Evaluate the combined effect of booking lead time and previous no-show history.

10. Overall Project Status

Week 6 successfully extended the HealthConnect project from exploratory analysis into deeper analytical investigation and decision support. New segment relationships were analysed, key findings were reviewed, the dashboard was enhanced, and practical recommendations were developed. Cross-track Data Science collaboration was attempted but could not be completed because a Data Science intern was unavailable. The resulting analysis and documented modelling requirements provide a clear foundation for future integration and Week 7 testing.